

Special Issue on Semantic Ontologies in Economics

Akshay Shanker and Christopher Carroll

July 30, 2026

Abstract

A computed model’s meaning is largely implicit in prose, notation, calibration, and code. Professional convention has helped fill the gaps. But economists now use large language models (AIs) to write, modify, and translate modeling code, and AI output is probabilistic by construction. Unverified, such code can compute a different model from the one intended, its results cannot be cross-checked or reused, and its economic interpretation becomes ambiguous. Verification requires a semantic ontology: an AI-readable, deterministic statement, independent of code and solver, of which economic objects exist, what stands for what, what is written down, and when the interpretation holds. We propose a special issue collecting semantic ontologies for modeling languages, toolkits, empirical methods, and model classes.

Society for Computational Economics Working Group No. 1 on Language and Formal Semantics.

1 Introduction and Proposal

When economists compute a model, its precise *meaning* is distributed, much of it implicitly, across prose, notation, calibration, and code. The paper says “we solve the following model”, but its equations underdetermine both what its code computes and what the computations mean, leaving the reader to fill the gaps. The same gaps

confront whoever rebuilds or extends the model. As a result, without a concrete statement of the model’s meaning to check against, computational results are hard to cross-verify, and interoperability between implementations becomes difficult.

So far, professional convention has helped trained modelers within a discipline to interpret, replicate, and build on one another’s models. But convention breaks down once AI is used to write, modify, and translate modeling code. To interpret or verify AI output we need a statement, independent of the code, of which economic objects are computed and which relations among them are enforced; we call such statements *semantic ontologies*. Without them, AI use is prone to error and misunderstanding. With them, translating models between toolkits, and even generating new modeling research, become operations a language model can carry out and be checked on cheaply, rather than manual recoding.¹

PROPOSED SPECIAL ISSUE

We propose a special issue that collects semantic ontologies for the domains of computational economics: modeling languages, toolkits, empirical methods, and model classes. Each paper takes one domain of study, states its semantic ontology, and demonstrates it on worked examples of the team’s own choosing. The SCE working group will submit a subset of the papers, an open call will invite the rest, and a comparison paper, written jointly by the participating teams, will close the issue.

1.1 Semantic Ontologies

Semantic ontologies are broadly defined as a formal way to codify the meaning of computational models. For our purposes, a semantic ontology consists of the objects and relations within the model, the meaning of those objects and relations, and the written forms that record them (syntax: a file, a model write-up, a specification).² Theory supplies part of the semantic ontology — a general-equilibrium model’s objects are precisely defined — but the semantic ontology must collect these definitions, map them concretely to computational and written counterparts, and state the

¹The nearest precedent in economics, the Macroeconomic Model Data Base (Wieland et al. 2012), compares models under common variables, common shocks, and a menu of common policy rules while each model keeps its own equations; it standardizes comparison, not meaning.

²In a heterogeneous-agent model, for instance, the ontology contains the response of decision rules to prices, the cross-sectional distribution decisions induce, and the feedback of that distribution into prices — as relations of the theory and as their computational counterparts.

assumptions under which the mapping holds. Without a written semantic ontology, nothing says which of the theory’s objects a given file or function call stands for, and ordinary solver code gives one executable realization, not a solver-independent statement of what the representation denotes.

Economics is not the first field to need a semantic ontology: manufacturing’s Process Specification Language (Grüninger and Menzel 2003; Bock and Grüninger 2005), Modelica for physical systems (Fritzson and Engelson 1998; Modelica Association 2023), planning’s PDDL2.1 (Fox and Long 2003), and neuroscience’s NeuroML (Gleeson et al. 2010) each attach an explicit, solver-independent meaning to a model representation. Industry has met the same need from the opposite direction, reconstructing never-written conceptual models of the business from its data and business processes (AWS Database Blog 2026). Compared with all of these, economics is well placed, because the relations among a model’s objects are the theory itself — stated in the paper, but not systematically attached to the files and function calls that compute it.

2 Guidelines for Papers

2.1 Research Domains and Themes

Domains of study. Each paper in the special issue will address one *domain of study*. Domains of study can include a modeling language, a toolkit, an empirical method, a model class, or a combination of these.

DOMAINS OF STUDY
• Modeling languages. A modeling language is a fixed grammar in which a complete model is written as a file, as in Dynare’s model language or Dolo’s YAML model files. • Toolkits. A toolkit is a collection of construction calls and classes from which a model is assembled in a programming language, as in HARK’s agent classes, the sequence-space Jacobian toolkit (Auclert et al. 2021), or the VFI Toolkit. • Estimation and empirical methods. An empirical method takes a model to data: it states which measured objects stand for the model’s quantities, prices, and shocks, which classifications organize them, and which transformations construct the inputs

to estimation or calibration, as in the simulated method of moments, indirect inference, or the social accounting matrices and input-output tables to which computable general-equilibrium models are calibrated.

- **Model classes.** A model class is a family of models built from the same kinds of objects, as in heterogeneous-agent macroeconomies or overlapping-generations economies.

Whichever domain of study a paper addresses, its theory and mathematics are taken as pre-existing: the Bellman equation is not in question, while what a given Dynare file or HARK model means is. For a language, a toolkit, or an empirical method, the semantic ontology may therefore emphasize the map from computational representations and procedures to the mathematical objects they stand for. A model-class domain of study, by contrast, may place greater emphasis on the objects and relations of the mathematics itself. Where the stated ontologies differ across papers, the difference is itself a result for the closing comparison to report.

Research themes. The research themes of the issue are the two tasks every paper shares: first, constructing a semantic ontology for its domain of study, and second, establishing the ontology’s *metatheory*, statements about the ontology rather than components of it. There are three components to each semantic ontology: what is assumed to exist (the *ontology*), what stands for and relates to what (the *denotation*), and what is written down (the *syntax*). Domains of study will differ in how important these components are, and some, such as a model class, may not have certain components, such as syntax. The metatheory states when the interpretation holds (*well-posedness*) and what preserves it (*equivalence*, *adequacy*, and *convergence*). Details of these research themes are given in Appendix A.

2.2 Methodology

Formalizing the semantic ontology. A requirement for each paper will be that it presents a *formal* account of the semantic ontology of its domain of study. However, the researchers choose the formalism in which they state these components.

Broadly, a team can choose among three families of formalism, which differ in how much meaning they fix (Uschold and Grüninger 1996). The first family, *mathematical semantics*, translates representations into mathematical structures and proves properties of the translation: a team may assign each written form an object

in the ordered structures of domain theory (denotational semantics; Scott and Strachey 1971), construct meaning as the unique homomorphism from a many-sorted algebra of terms (initial-algebra semantics; Goguen, Thatcher, Wagner and Wright 1977), interpret a typed syntax in a category with matching structure (typed categories; Lambek and Scott 1986), write rules specifying execution on an abstract machine (operational semantics; Plotkin 1981), or state the assertions that hold before and after execution (axiomatic semantics; Hoare 1969).

Where mathematical semantics builds a map for the domain’s own syntax, the second family, *logical axiomatization*, writes the ontology as axioms in an existing formal logic — in current practice a description logic — and the logic’s ready-made model theory then fixes the admissible interpretations (Gruber 1993; Baader et al. 2017).

Where logical axiomatization states a theory, the third family, *graphs and diagrams*, records only the entities and relations — an entity-relationship diagram (Chen 1976), a UML class diagram (Berardi, Calvanese and De Giacomo 2005), or a knowledge graph (Hogan et al. 2021) — with little or no axiomatization. The last two families overlap, since a knowledge graph written in a web ontology language is at once a graph and a logical theory.

Different fields have made different choices: programming-language researchers state meaning mathematically, biomedical consortia axiomatize their ontologies in description logic (Ashburner et al. 2000), and industry records knowledge graphs from observed data, without axioms (Noy et al. 2019).

Which family fits a paper depends on whether its domain of study has a syntax. A team whose domain has one — a modeling language or a toolkit — can use any of the mathematical semantics; a team whose ontology has no syntax — a model class, or an empirical method before its specification is constructed — can axiomatize the ontology in a logic or record it as a graph. A team seeking machine-checked guarantees can go further and formalize its domain in a proof assistant such as Lean, defining the model objects as types and the file-to-object map as a function inside the system, so that well-posedness and equivalence become theorems the machine verifies (de Moura and Ullrich 2021).

Top-down vs. bottom-up approaches. A team can also build its semantic ontology top-down or bottom-up. Building top-down, it starts from the theory and formalizes the theory’s objects; building bottom-up, it starts from observed data and relations and abstracts the ontology from them, as industry does when it reconstructs

a conceptual model from a data catalog (AWS Database Blog 2026). In this issue we expect economic theory to supply the structure, so most papers will work top-down; a bottom-up construction may suit agent-based modeling or policy research, where observed relations precede a settled theory.

3 Organization

Papers pass an internal review within the working group [process TBC] and then the journal’s ordinary external refereeing. We expect six to eight papers, with a session at the Society’s conference between submission and revision.

3.1 Teams and Editors

The working group’s members include developers of Dynare, HARK, and the VFI Toolkit. Committed teams: [to be listed — only teams that have agreed in writing]. The working group’s papers will come from its subgroups. Akshay Shanker and Matthew McKay chair the working group and coordinate the teams and the internal review. The guest editors (a lead editor and at least one co-editor, neither submitting to the issue) are drawn from outside the working group [names, institutions — TBC]. Papers are submitted through the journal’s editorial system, marked for this special issue, and are refereed under the journal’s ordinary standards; the guest editors handle every paper, including the comparison paper, subject to the journal’s final editorial authority.

3.2 Timeline

Calendar dates TBC on acceptance.

- Call for papers [on acceptance]
- Submissions [+9 months]
- Internal review completed [+12 months; process TBC]
- Referee reports [+15 months]
- Revisions and the comparison paper [+24 months]

4 References

Ashburner, M., C. A. Ball, J. A. Blake, et al. (2000). “Gene Ontology: tool for the unification of biology.” *Nature Genetics* 25(1), 25–29. [\[link\]](#)

Auclert, A., B. Bardóczy, M. Rognlie, and L. Straub (2021). “Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models.” *Econometrica* 89(5), 2375–2408. [\[link\]](#)

AWS Database Blog (2026). “Build a semantic ontology to power AI assistants on AWS — Part 1.” 14 July 2026. Accessed 29 July 2026. [\[link\]](#)

Baader, F., I. Horrocks, C. Lutz, and U. Sattler (2017). *An Introduction to Description Logic*. Cambridge University Press. [\[link\]](#)

Berardi, D., D. Calvanese, and G. De Giacomo (2005). “Reasoning on UML Class Diagrams.” *Artificial Intelligence* 168(1–2), 70–118. [\[link\]](#)

Bock, C., and M. Grüninger (2005). “PSL: A Semantic Domain for Flow Models.” *Software and Systems Modeling* 4(2), 209–231. [\[link\]](#)

Chen, P. P. (1976). “The Entity-Relationship Model — Toward a Unified View of Data.” *ACM Transactions on Database Systems* 1(1), 9–36. doi:10.1145/320434.320440

de Moura, L., and S. Ullrich (2021). “The Lean 4 Theorem Prover and Programming Language.” In *Automated Deduction — CADE 28*, Lecture Notes in Artificial Intelligence 12699. Springer, 625–635. doi:10.1007/978-3-030-79876-5_37

Fox, M., and D. Long (2003). “PDDL2.1: An Extension to PDDL for Expressing Temporal Planning Domains.” *Journal of Artificial Intelligence Research* 20, 61–124. [\[link\]](#)

Fritzson, P., and V. Engelson (1998). “Modelica — A Unified Object-Oriented Language for System Modeling and Simulation.” In *ECOOP ’98 — Object-Oriented Programming*, Lecture Notes in Computer Science 1445. Springer, 67–90. doi:10.1007/BFb0054087

Gleeson, P., S. Crook, R. C. Cannon, M. L. Hines, G. O. Billings, et al. (2010). “NeuroML: A Language for Describing Data Driven Models of Neurons and Networks with a High Degree of Biological Detail.” *PLoS Computational Biology* 6(6), e1000815. doi:10.1371/journal.pcbi.1000815

Goguen, J. A., J. W. Thatcher, E. G. Wagner, and J. B. Wright (1977). “Initial Algebra Semantics and Continuous Algebras.” *Journal of the ACM* 24(1), 68–95. doi:10.1145/321992.321997

Gruber, T. R. (1993). “A translation approach to portable ontology specifications.” *Knowledge Acquisition* 5(2), 199–220. [\[link\]](#)

Guarino, N., D. Oberle, and S. Staab (2009). “What Is an Ontology?” In S. Staab and R. Studer (eds.), *Handbook on Ontologies*, 2nd ed. Springer, 1–17. [\[link\]](#)

Grüninger, M., and C. Menzel (2003). “The Process Specification Language (PSL): Theory and Applications.” *AI Magazine* 24(3), 63–74. doi:10.1609/aimag.v24i3.1719

Harel, D., and B. Rumpe (2004). “Meaningful modeling: what’s the semantics of ‘semantics’?” *Computer* 37(10), 64–72. [\[link\]](#)

Hoare, C. A. R. (1969). “An axiomatic basis for computer programming.” *Communications of the ACM* 12(10), 576–580. doi:10.1145/363235.363259

Hogan, A., E. Blomqvist, M. Cochez, C. d’Amato, G. de Melo, et al. (2021). “Knowledge Graphs.” *ACM Computing Surveys* 54(4), article 71. doi:10.1145/3447772

Lambek, J., and P. J. Scott (1986). *Introduction to Higher Order Categorical Logic*. Cambridge University Press. [\[link\]](#)

Modelica Association (2023). *Modelica Language Specification*, version 3.6. [\[link\]](#)

Noy, N., Y. Gao, A. Jain, A. Narayanan, A. Patterson, and J. Taylor (2019). “Industry-scale knowledge graphs: lessons and challenges.” *Communications of the ACM* 62(8), 36–43. [\[link\]](#)

Plotkin, G. D. (1981). *A Structural Approach to Operational Semantics*. Report DAIMI FN-19, Computer Science Department, Aarhus University.

Scott, D., and C. Strachey (1971). “Toward a mathematical semantics for computer languages.” *Proceedings of the Symposium on Computers and Automata*, Polytechnic Institute of Brooklyn, 19–46. [\[link\]](#)

Uschold, M., and M. Grüninger (1996). “Ontologies: principles, methods and applications.” *The Knowledge Engineering Review* 11(2), 93–136. [\[link\]](#)

Wieland, V., T. Cwik, G. J. Müller, S. Schmidt, and M. Wolters (2012). “A new comparative approach to macroeconomic modeling and policy analysis.” *Journal of Economic Behavior & Organization* 83(3), 523–541. doi:10.1016/j.jebo.2012.01.006

A Research Themes

A paper addresses those of the following five research themes relevant to its domain of study. The first three are the components of the domain’s semantic ontology; the last two are statements about it — when the interpretation holds, and what preserves it under changes of representation and computation:

THE RESEARCH THEMES

1. **Ontology: what is assumed to exist.** The economic and mathematical entities and relations within the domain — agents, states, shocks, timing, the equilibrium concept, operators, functions, and which objects determine which — together with the criteria for when two of them are the same (Gruber 1993; Guarino, Oberle and Staab 2009). The ontology is stated in its own terms, without reference to the syntax.
2. **Denotation: what stands for what.** The meaning of the ontology’s objects. The meaning takes the form of relationships between the mathematical, economic and computational representations — for instance, value functions as economic concepts, instantiated on a mathematical space, and approximated on the computer. Typically the denotations will also include relationships between these representations and the written forms: each written form stands for its mathematical or economic object under an explicit map (Harel and Rumpe 2004).
3. **Syntax: what is written down.** The set of legal written forms: the model file, the sequence of construction calls, or the specification of an estimation exercise. For instance, a language has a full grammar; a toolkit may expose construction calls or classes. A domain (such as an empirical method) may have no syntactic written form at all, in which case much of the role of syntax is played by the denotation.

The next two research themes are not components of the semantic ontology; they are statements about it:

4. **Well-posedness: when the interpretation holds.** The conditions under which the denotation is well defined: domains, units, timing, information structure, and the parameter restrictions assumed.
5. **Equivalence, adequacy, convergence: what preserves it.** Denotational equivalence: two written forms, one denotation. Adequacy: an implementation computes exactly the object denoted. Convergence: a numerical approximation approaches it as grids and tolerances are refined.